Incident forecast: NET-062

Situation as of **Sat 31 Oct 2026 · 15:02:42** · stage 2 of 3 (1 hop observed) · 73 machines, 68 users

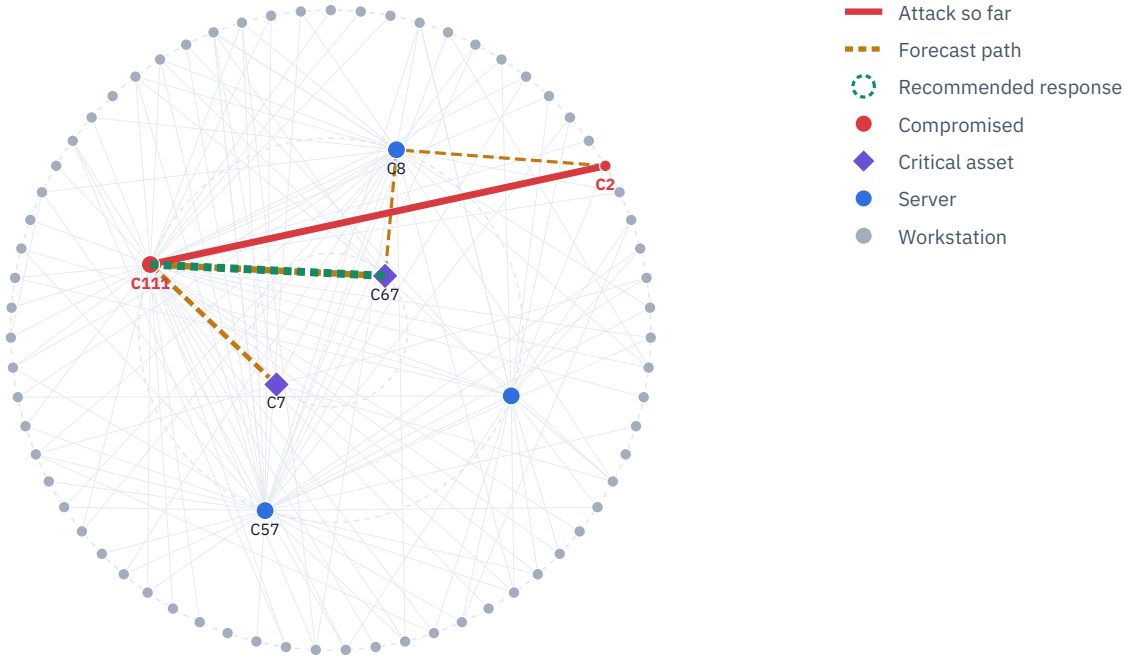
Compromised machines	Attacker is on	Risk to critical assets	Forecast confidence
2	**C111**	**28%**	**33%**

RECOMMENDED NEXT CUT

Revoke access from C111 to C67

Removes **25%** of the forecast path risk while blocking **0.0%** of normal machine-to-machine traffic from the last 24 hours. Chosen by the defense simulator as the action with the best risk removed minus traffic blocked, among actions beyond the machines already known to be compromised. Standard containment, isolating already-compromised C111, removes **60%** of the risk and blocks **26.1%** of traffic.

Network view



Forecast: attacker's next move

Calibrated probability that the attacker on **C111** moves to each machine next. Uncertainty is the binary entropy of that probability (0 sure, 1 coin flip).

#	Machine	Type	Probability	Uncertainty
1	C57	Web server	16.2%	0.64
2	C8	App server	14.2%	0.59
3	C67	Domain controller	11.6%	0.52
4	C65	Print server	10.7%	0.49
5	C113	Workstation	8.0%	0.40

Most likely paths to critical assets

Ranked with Yen's k-shortest paths on move costs of −log p. A path's probability is the product of its hops.

#	Route	Hops	Probability
1	C111 → C67	1	0.116
2	C111 → C7	1	0.076
3	C2 → C8 → C67	2	0.050
4	C111 → C57 → C67	2	0.020
5	C111 → C57 → C7	2	0.014

Other responses considered

Alternative response	Risk removed	Traffic blocked	Utility
Revoke C111 → C7	15%	0.0%	0.15
Revoke C8 → C67	4%	0.7%	0.03
Revoke C57 → C67	3%	0.3%	0.03
Revoke C65 → C67	1%	0.0%	0.01
Revoke C113 → C67	1%	0.3%	0.00

Flagged log events

Logins from compromised machines, checked against the forecast made before they happened. HIGH: destination was in the top-3 forecast. MEDIUM: any other login from a compromised machine, or a very unexpected connection.

Time	User	Connection	Flag	Forecast rank	Unexpected
2026-10-31 15:02:42	U23	C2 → C111	HIGH	#2	0.02
2026-10-31 12:20:53	U46	C11 → C89	MEDIUM	-	0.97
2026-10-31 10:41:27	U31	C102 → C40	MEDIUM	-	0.96
2026-10-31 10:23:23	U66	C63 → C10	MEDIUM	-	0.97
2026-10-30 15:45:06	U20	C107 → C19	MEDIUM	-	0.96

How reliable is this forecast?

Measure (45 unseen test networks)	Temporal GNN	Rule-based	Random
Next hop in top 5	70%	66%	9%
Next hop in top 1	24%	18%	2%
Attack caught before target	73%	56%	-
Calibration error (ECE)	0.095	0.198	-
Next-hour link AUC	0.970	-	-

Appendix: ground truth (evaluation only)

Synthetic networks record where the attacker really went. A real deployment would not have this section.

Item	Value
Real remaining path	`C111 → C57 → C67`
Real next machine	C57 (forecast rank #1)
Real path among top-5 forecast paths	Yes